

Table 1: MongoDB vs RethinkDB vs SQLite vs Apache Cassandra

Criteria	MongoDB	RethinkDB	SQLite	Apache Cassandra
Latest version	MongoDB 6.0	RethinkDB 2.4.2	SQLite 3.42.0	Apache Cassandra 4.1
Released in	2009	2012	2000	2008
Supportable programming languages	C++, Python, JavaScript	C++, Python, JavaScript	Python, Java, C#, Ruby, PHP	C++, Java, Python, Go, Node.JS
Database model	Supports JSON	Supports JSON	Follows a relational data model and supports SQL queries.	Follows NoSQL database
Key features	Autosharing, indexing	Distributed database, automatic sharding, and replication	Serverless, flexible	Easy data distribution, fast linear-scale performance
Advantages of using in IoT applications	Document-oriented, flexible schema	Adaptable query language, plug-and-play functions	Enables small memory footprint, no dependencies	Fault-tolerant, scalable

Comparison of MongoDB, RethinkDB, SQLite, and Apache Cassandra

Popular databases used in IoT applications are MongoDB, RethinkDB, SQLite, and Apache Cassandra. The choice between these depends on specific project requirements. MongoDB and RethinkDB are document databases suitable for handling diverse and evolving data, with MongoDB having broader industry adoption. SQLite is a lightweight embedded database for local storage, while Cassandra excels in high scalability, availability, and distributed architectures. Table 1 compares these four databases critically.

Installation of MongoDB on Windows

MongoDB 6.0 is the latest community edition version and can be downloaded on the official URL of MongoDB. To install MongoDB on Windows, a 64-bit system is required. The steps are given below.

Step 1: Download the MongoDB community .msi installer zip package and unpack it in your system directory.

Step 2: Next, navigate to the download directory; there should be an installer file called .msi. From the admin user, the installation wizard will be launched by double-clicking the .msi file, as shown in Figure 2.

Step 3: After running the .msi installation file of MongoDB, choose the complete setup type. Configure and start MongoDB as a Windows service during the installation, and the MongoDB service will start on successful installation, as shown in Figure 3.

Step 4: Now, the user can edit the system environment variable from the 'program files' of the C drive.

Step 5: The MongoDB services can now be accessed at the local host, as shown in Figure 4.

Popular commands used in MongoDB for IoT

MongoDB offers a rich set of commands that can be useful

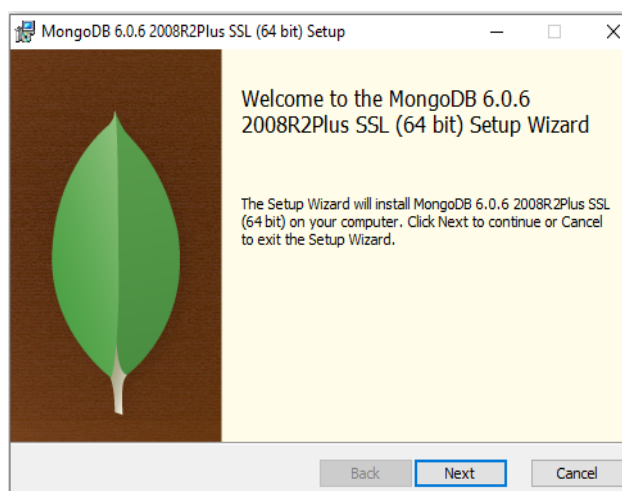


Figure 2: Setup wizard of MongoDB installation

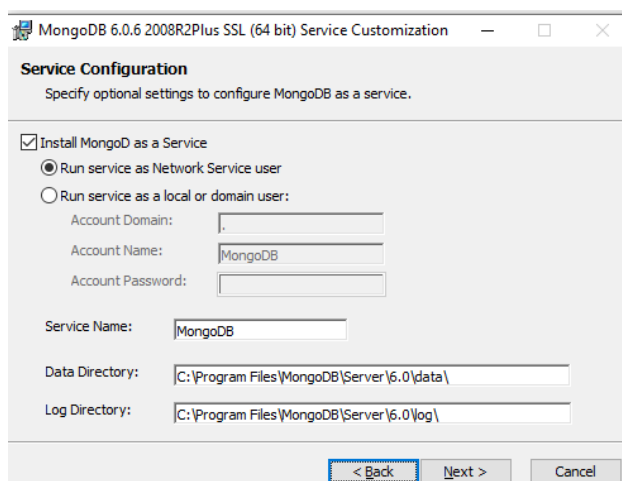


Figure 3: MongoDB service customisation